VISHWAS MAMGAIN

Java Backend Engineer | Spring Boot | Microservices | Fintech & AI Systems

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PROFESSIONAL SUMMARY

Java Backend Engineer with 4+ years of experience designing, developing, and scaling backend systems using Java 17, Spring Boot, Spring MVC, and microservices architecture. Proven track record building secure, high-throughput REST APIs (Spring Security, JWT, RBAC), event-driven systems with Apache Kafka (Retries, Idempotency, Dead-Letter Queues), and Redis-based caching layers — processing 10,000+ fintech transactions/day. Built end-to-end fintech portfolio management systems spanning digital gold, unlisted stocks, mutual funds, PMS, and liquiloan, including third-party API integrations (MMTC-PAMP), FIFO-based profit & loss logic, scheduled valuation jobs, and complex multi-table aggregation APIs. Reduced AI API costs by 70% using Semantic Caching and improved backend latency by 30% through Redis caching and SQL optimization. Currently building an AI-powered Job Tracker using Spring Boot, Spring AI, Redis, JWT, and Docker. Strong foundation in relational databases (PostgreSQL, MySQL), containerization (Docker, Kubernetes), CI/CD (Azure DevOps), and testing (JUnit 5, Mockito). Comfortable working independently or within cross-functional, Agile/Scrum, and remote/distributed teams.

CORE SKILLS

Languages & Core: Java 17, Object-Oriented Programming (OOP), Data Structures, REST API Design

Frameworks & Backend: Spring Boot, Spring MVC, Spring Security, Spring Batch, Spring AI, Hibernate, Microservices, Event-Driven Architecture

Security: JWT Authentication, RBAC, API Security Best Practices

Messaging & Async Processing: Apache Kafka, Retries, Idempotency, Dead-Letter Queue (DLQ), Asynchronous Workflows

Databases & Performance: PostgreSQL, MySQL, Redis, SQL Optimization, Query Tuning, Indexing, Complex Multi-Table Aggregation Queries

Fintech & Portfolio Systems: Multi-Asset Portfolio Management (Digital Gold, Unlisted Stocks, Mutual Funds, PMS, Liquiloan), FIFO-Based P&L Logic, Cron-Based Valuation Jobs, Third-Party API Integration (MMTC-PAMP)

AI & Integrations: Spring AI, OpenAI/Groq LLM APIs, Semantic Caching, Confidence-Based Routing

DevOps & Tools: Docker, Kubernetes, Azure DevOps, Git, Maven, CI/CD Pipelines

Testing & Quality: JUnit 5, Mockito, Postman, Code Reviews, Debugging & Troubleshooting

Methodologies: Agile/Scrum, Software Development Lifecycle (SDLC), Cross-Functional & Remote Collaboration

PROFESSIONAL EXPERIENCE

Freelance Backend Engineer | Self-Employed *April 2026 – Present*

- Designed, developed, and maintained a scalable AI-Powered Job Tracker end-to-end using Java, Spring Boot, PostgreSQL, Redis, Spring AI, JWT, and Docker Compose, following SDLC best practices.
- Integrated Spring AI with a third-party Groq LLM API to deliver natural language search across structured job data.
- Implemented Redis caching with TTL-based invalidation to optimize application performance and reduce redundant database queries.
- Containerized the application with Docker Compose for consistent local development and one-command deployment.

Software Engineer | FINCART | Noida *March 2024 – April 2026*

- Owned end-to-end portfolio management APIs across multiple asset classes — Digital Gold, Unlisted Stocks, Mutual Funds, PMS, and Liquiloan — covering buy/sell order flows and real-time portfolio valuation for customers.
- Built the digital gold purchase journey by integrating third-party MMTC-PAMP APIs, handling real-time buy/sell transactions end-to-end.
- Designed and implemented FIFO-based portfolio accounting logic to accurately compute holdings and realized/unrealized profit & loss on every buy/sell transaction.
- Built and scheduled cron jobs to refresh daily portfolio valuations across asset classes based on latest market prices (e.g., daily gold rates), ensuring customers always saw an accurate, up-to-date portfolio.
- Designed complex, high-performance GET APIs aggregating data across multiple database tables to power varied portfolio views, filters, and sorting scenarios.

- Designed Kafka-based asynchronous workflows to decouple transaction ingestion from downstream processing, improving system resilience.
- Reduced AI API costs by 70% by designing a semantic caching layer with confidence-based request reuse.
- Improved backend response latency by 30% through Redis caching and SQL query optimization, supporting 10,000+ transactions/day.
- Implemented secure REST APIs using Spring Security, JWT Authentication, and RBAC; automated reconciliation workflows using Spring Batch and contributed to CI/CD pipelines in Azure DevOps, reducing manual effort by 40%.

Junior Software Engineer | ARJH Tech Labs | Noida *September 2021 – March 2024*

- Built and maintained RESTful APIs using Java, Spring Boot, Hibernate, and PostgreSQL, and migrated monolithic applications to a microservices architecture.
- Designed fault-tolerant Kafka consumers using Retries, Idempotency, and Dead-Letter Queues (DLQ) to ensure reliable, scalable data processing.
- Optimized 15+ SQL queries for efficient data management, reducing execution time from ~2 seconds to ~200 ms (~90% improvement).
- Debugged and resolved production issues involving memory leaks, thread contention, and cache inconsistencies, improving application stability and performance.

PROJECTS

AI-Powered Transaction Classification Platform

Java | Spring Boot | Apache Kafka | Redis | PostgreSQL | Spring AI | GPT APIs | Microsoft Azure

- Developed an event-driven backend for automated fintech transaction classification, processing 10,000+ transactions/day.
- Reduced GPT API calls by 70% using Redis semantic caching and implemented confidence-based routing for automated classification.
- Improved system reliability using retries, idempotent consumers, and dead-letter queues (DLQ).

EDUCATION

Bachelor of Technology (B.Tech) — Swami Rama Himalayan University | 2019